

Is this the right time for Sudan to replace diesel-powered generator systems with wind turbines?



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ABSTRACT

Access to electricity is vital in rural regions of Africa. There is substantial potential to utilise wind energy, yet investment in renewable energy projects is trivial because it has been insufficiently endorsed. As a result, most people living in remote regions of Africa rely entirely on diesel-powered generators, which expel pollutants in substantial amounts. These adversely affect the health of people and the environment. Sudan – one of Africa's largest countries – has a range of resources from which renewable energy could be generated, including favourable wind power generating conditions. This paper represents the first effort in literature to use a strategic perspective to explore how viable wind energy systems are in Sudan. It reports a study using the Hybrid Optimization Model for Electric Renewables (HOMER) across 21 diverse sites in Sudan. It identifies the optimal wind turbine system and best locations and analyses the costs involved and how much pollution could be prevented by implementing wind farms instead of diesel systems. The results establish that the optimal type of wind turbine across the investigated locations is the Enercon E-138 EP3. Wind farm use is expected to avoid almost 49 million kg/year of pollutants and reduce energy cost by US\$0.91317/kWh. The optimal sites for utilising wind energy were Wawa, Dongola, Karima, Wadi Halfa and Al-Goled in order of suitability and their low costs of electricity and high wind speed records. Wind turbine pricing is predicted to reduce over time, so the study varied the cost of wind turbines to understand the impact on project economics. A reduction in wind turbine costs by 10 % lowered energy cost to between US\$0.03681/kWh and US\$0.04052/kWh. A further 25 % price drop reduced the cost of energy to between US\$0.03067/kWh and US\$0.03376/kWh. This study reveals the significant potential of integrating renewable energy into the energy mix of Sudan and provides an essential reference for wind farms in the energy sector across other developing countries.

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1. Introduction

The quest to utilise renewable energy (RE) resources to generate energy has become vital. Almost 1.3 billion globally cannot access modern sources of energy [1]. Many countries have expanded the development of RE resources in their territories and have developed complementary manufacturing industries [2]. The distinct advantages of RE over other conventional energy sources, including the implications of climate change, have spurred targets and incentives aimed at developing RE. Limited time and the failure to date of agreements made in the conference of parties (COP25) to

suppress CO₂ emissions by 2050 [3] have created the need for radical policies to promote and support a massive inclusion of RE projects into the market.

As documented by the World Wind Energy Association (WWEA), wind energy potential is substantial and estimated to comprise at least 95,000 GW [4]. Yet, the global installed capacity is merely 650 GW [5]; wind energy is therefore an untapped source of energy. The installed capacity of wind energy technology grew by 60 GW – almost 20 % – in 2019, which represented its second substantial annual increase in the power market. This was due to policy changes, mainly in the USA and China [6]. Even though REN21 estimated that by the end of 2019, wind technology commercialisation would reach 102 countries, only 5 % of global energy production was generated from wind energy in 2019. Thus, it is imperative to pave the way for a more comprehensive inclusion of wind technology in the installed capacity of countries endowed

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with a high potential for wind resources.

More than 600 million people in Africa do not have access to electricity. Several African countries facing rapid economic growth are challenged by an alarming increase in energy demand [7]. In most rural areas, transmission line extensions and diesel generators dominate the electricity supply system [8]. Africa is considered to be a continent blessed with RE resources [7], but harvesting these resources to reach the goals stated in several countries' roadmaps is slow [9]. In this regard, there is a need to carefully assess and investigate the implementation of RE technologies such as wind and solar energy, which are typically land-dependent [7]. assessed wind energy potential in Africa using a geographic information systems approach, and the results show that several African countries (Sudan, Algeria, South Africa, Morocco, etc.) possess high potential for wind power. Furthermore, they investigated the most suitable locations for wind energy projects. Fig. 1 shows each country's suitability with or without the national grid restriction and reveals that many African countries have the potential to implement wind technology projects to support their national installed capacity.

Gabra et al. [10] utilised a spatial mapping model to compare the cost-effectiveness of wind, solar photovoltaic (PV) and diesel generator systems. Their results showed that wind energy technology was more cost-effective in areas where the local average wind velocity is high. For the remaining locations, PV and diesel were more economical. However, the authors advised that supplementary information is needed to select technologies in Africa because of the considerable uncertainties assumed during their analysis. Given the above, regional or local feasibility studies are advisable. These should consider the relevant political framework regarding incentives and updated cost in line with their relative contribution to the existing energy mix. It is evident that significant potential exists across several countries, including Sudan, Djibouti and Nigeria [7,11,12].

Sudan is located between latitudes 3° N and 23° N, and longitudes 21° 45' E and 39° E, and is one of the largest countries of the African continent. Almost 70 % of the population lack access to electricity, and of the total national energy supply, biomass accounts for 62 % and fossil fuels 34 % [13]. Only 30 % of people who live in cities can access the electricity grid; 70 % of people who live in remote areas rely entirely on generators fuelled by diesel for

their daily electricity needs [14]. Diesel-based systems, however, are noisy, polluting and expensive to maintain [15]. The predominance of diesel generators in rural areas reflects insufficient investment into the national electricity grid, leading to sparse coverage and an intermittent supply.

To address these inadequacies, the government of Sudan has drawn a roadmap to improve the power sector's condition. It outlines a plan to increase the energy share from renewables from 0.38 % to 20 % by 2030 and 100 % by 2050 [16]. Of this planned capacity, wind power is intended to occupy the most significant share (680 MW; MW) of the total targeted installed capacity by 2031; this reflects the fact that Sudan is endowed with a high wind potential [16]. This progress is possible because the USA is expected to end its economic sanctions, making Sudan eligible to access support for debt relief provided through the International Monetary Fund [17]. This, along with the government's transformative roadmap, provides the impetus to improve the power sector in the coming years.

Before the implementation of any RE system, a thorough feasibility study is crucial to provide decision-makers with data that provides confidence in the process. If RE systems are improperly designed or oversized, the installation costs are either over or underestimated, with considerable consequences. Therefore, a techno-economic investigation is vital to correctly determine in advance the optimal scale and nature of the critical factors for a project. Several software tools have been created to assist practitioners in predicting and analysing the variables that will optimise a planned RE system. Many of these tools have shown significant potential, particularly in Africa; see Refs. [18,19].

Many software tools allow practitioners to design and plan RE systems. As documented by Sinha and Chandel [18], the Hybrid Optimization of Multiple Electric Renewables (HOMER) has stood out distinctively as the most integrated and powerful tool. It is widely recommended per global standards for optimal planning and design of RE systems. HySim, Dymola/Modelica, HYBRIDS, HYBRID DESIGNER, SOMES, Hybrid2, RETScreen, INSEL, iGRHYSO, IPSYS, iHOGA, ARES, RAPSIM, TRNSYS, SOLSIM, iGRHYSO, SOLSTOR, HySys are amongst the 19 tools assessed in this study. HOMER maintains a reasonably high level of accuracy when compared to traditional statistical methods [20,21], frequency-domain methods [22,23] and simplified linear programs over limited time periods

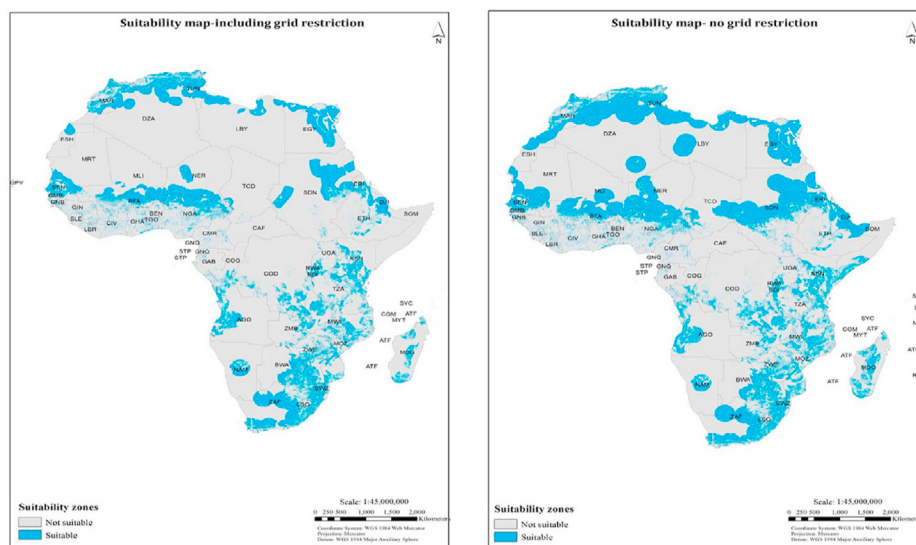


Fig. 1. Suitability map for wind power installations in Africa [7].

[24–26]. HOMER's hourly-based calculations provide improved accuracy compared to the aforementioned existing methods that typically evaluate the average monthly performance of a system [27]. The US Department of Energy's National Renewable Energy Laboratory developed HOMER, and since 1992 it has become gold-standard software globally for decision-making for microgrid and distributed energy resource.

HOMER can accurately simulate and optimise power systems, both stand-alone and grid-connected, with RE technologies including wind, PV, biomass and hydropower. It facilitates quick feasibility studies, which are essential for the design phase of most projects and often central to deploying RE technologies. The software can be customised for different scenarios and equipment types. It has been widely applied, for example, in Africa [19]; the USA [28]; Turkey [29]; Germany [30]; Algeria [31]; and Greece [32].

To see how feasible it would be to implement RE projects in Africa to overcome various challenges facing the energy sector, Nacer et al. [33] investigated the feasibility of grid-connected RE (wind and PV) systems for a typical farm in Algeria. First, an assessment of wind and solar resources across the country was conducted. A techno-economic sensitivity analysis was then performed to determine the best configuration for the selected locations. The simulations showed that for the northern part of the country, solar PV systems were more suitable. However, wind technology performed better in the highlands, where the wind potential is significant. It was found that these systems would increase the reliability of the national grid and reduce CO₂ emissions compared to conventional methods.

Ayodele et al. [34] investigated the feasibility of a hybrid wind, solar and diesel system, including storage batteries, in a village in Kwara State, Nigeria. A micro bank's energy demand was first assessed to design the appropriate energy system configuration. Subsequently, the authors performed a techno-economic sensitivity analysis to optimise the hybrid system. The results showed that HOMER software delivered an optimally designed hybrid system, in contrast to an analytical method overestimating the cost. The study reported a levelised electricity cost of US\$0.667/kWh with a reduction of almost 50 % of CO₂ emissions.

In South Africa, Adefarati and Obikoya [35] conducted a techno-economic investigation into the implementation of wind turbines. They firstly assessed the wind energy resource, including the load in different locations. HOMER software was then used to identify and evaluate the techno-economic performance of a grid-connected wind turbine system in the selected sites, taking into account the government roadmap and planned power sector reform. The economic analysis results on the net present cost (NPC) and cost of energy (COE) showed that Cape Town could deliver the best performance of all the selected locations, based on its high average wind speed of 6.39 m/s.

Azerefegn et al. [36] used HOMER to investigate the feasibility of a hybrid wind-PV system in three climate regions in Ethiopia. Because Ethiopia is known for its weak power infrastructure, four scenarios of unscheduled electricity outages were considered in the study. The results of the designed systems of approximately 50 kW concurred with that reported earlier. In regions with high average wind speed, wind technology was more feasible than PV-based systems. The COE in all three locations was lower than that reported for South Africa by Adefarati and Obikoya [35], probably because of differences in the economic parameters considered.

Motivated by the promising energy roadmap of the Sudanese government, Elkadeem et al. [15,37] presented a techno-economic analysis of a hybrid system isolated from the grid and designed to power irrigation and agriculture in Dongola, Sudan. Several configurations of wind, PV, battery storage, converter technologies and diesel generators were investigated. In order to identify the most

suitable configuration, the stochastic behaviour of RE technologies was considered. The simulation revealed that a combination of all previous technologies delivered the best economic performance compared to all remaining cases, with a COE of US\$0.387/kWh. Furthermore, integrating RE technologies contributed considerably toward reaching an energy excess of 9.65 %. Finally, the authors highlighted the significance of these studies toward achieving the United Nations Sustainable Development Goals.

Salih et al. [38] evaluated several hybrid systems designed to support the telecommunication equipment load in a remote area of Sudan. Three configurations were considered in the study: PV/wind turbine/diesel, generator/battery, and PV/diesel generator/battery. Results have shown that RE-based systems provide the reliability of 100 % throughout their 20-year lifespan. Thus, the study concluded that green energy technologies present an essential option for telecommunications systems in Sudan.

Also, in Sudan, this time in Khartoum, Abdallah et al. [39] investigated the feasibility of wind, PV and battery hybrid system. Different load profiles were considered, starting from a single home with a load of 338 Wh/day to 50 homes with a demand of 16.9 kW/day. The results showed that during the 25 years of the system's lifespan, increasing the number of houses decreased the COE by more than 50 %. Moreover, the study confirmed the need for government support to lower the cost of wind turbines to improve the viability of hybrid systems.

To shift energy generation to sustainable methods, considerable effort is being made to promote the global realisation of RE projects, particularly wind turbine-based projects. Still, there has been a dearth of research in Sudan to investigate the feasibility of these technologies. The investigations to date have been restricted to only a few sites rather than the entire country. Against this backdrop, this paper investigates the prospects of harvesting wind energy in Sudan to upscaling the country's energy outlook. Using data from 21 sites, simulations were carried out for a grid-connected wind farm power plant, focusing on identifying the best wind turbine technology and the most suitable location for wind energy generation. The effect of turbine price on project economics was assessed, as was the reduction in pollutant emissions resulting from implementing wind farm power plants instead of using diesel fuel.

The lack of modern electricity services stifles income-generating activities and hampers the provision of essential services such as health care and education. Therefore, the current work has a social nature as the study aims to take advantage of the resources available on the spot to generate electricity for people on the margins of society. Furthermore, the study contributes to the promotion and utilisation of renewable energies. This critical factor will contribute to achieving multiple criteria of the Sustainable Development Goals (SDGs), including economic, environmental, and social aspects. In the long run, this will lead to generating employment opportunities and reducing barriers and increasing acceptance of wind energy as an option to address the nation's long-term energy needs.

2. Materials and methods

2.1. Description of the study area

This investigation was undertaken to find the best type of wind turbine to employ to supply an electric load of 20 MW to each of 21 selected towns. Thus, using the HOMER software, the following diverse sites in Sudan were investigated: Wadi Halfa, Salala, Wawa, Port Sudan, Abu Hamed, Dongola, Karima, Merowe, Al-Goled, Haiya, Ad-Damar, Kassala, Omdurman, Al-Qadarif, Kutum, Al-Fashir, Al-Junaynah, Al-Obeid, Kosti, Al-Damazin, and Nyala. The aim was to pinpoint localities able to provide significant wind energy while also considering the COE. Details of sites' geography are

Table 1
Geographical information of the 21 locations under assessment.

No.	Location	Latitude (°N)	Longitude (°E)
1	Wadi Halfa	21.799	31.371
2	Salala	21.321	36.216
3	Wawa	20.444	30.351
4	Port Sudan	19.590	37.190
5	Abu Hamed	19.535	33.321
6	Dongola	19.146	30.470
7	Karima	18.532	31.827
8	Merowe	18.469	31.816
9	Al-Goled	18.471	30.644
10	Haiya	18.333	36.385
11	Ad-Damar	17.580	33.969
12	Kassala	15.458	36.404
13	Omdurman	15.647	32.481
14	Al-Qadarif	14.024	35.368
15	Kutum	14.203	24.664
16	Al-Fashir	13.620	25.355
17	Al-Junaynah	13.448	22.465
18	Al-Obeid	13.178	30.216
19	Kosti	13.123	32.650
20	Al-Damazin	11.785	34.342
21	Nyala	12.052	24.880

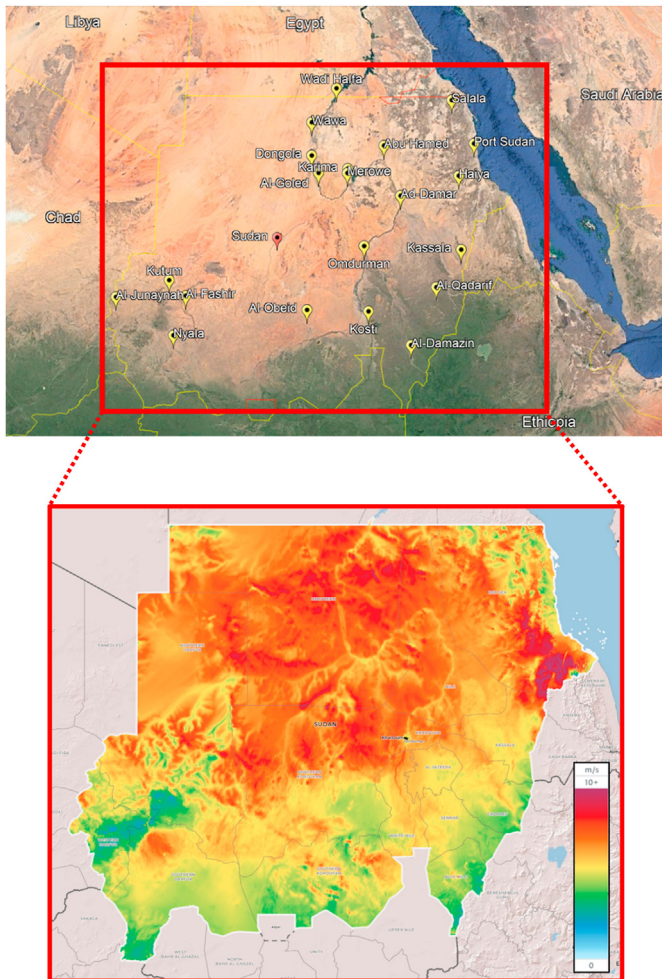


Fig. 2. The geographical sites under assessment for wind energy production in Sudan outlining the country's wind energy potential [40].

summarised in Table 1. Fig. 2 depicts the geographical location of each site, highlighting the county's wind energy potential.

2.2. Collection of input data and HOMER modelling framework operation

The preliminary stage of using the HOMER modelling framework was to acquire a range of input data for the domain of the study (i.e. load profile, meteorological data, replacement, capital, and operational and maintenance (O&M) expenses). In this investigation, 20 MW was assumed as the load profile for all months of the year. Twenty-five diverse types of wind turbine systems with a capacity ranging from a minimum of 1000 kW (1 MW) to a maximum of 7580 kW (7.58 MW) were selected from the HOMER modelling framework database and assessed independently.

Given the lack of data on replacement, capital, and O&M cost associated with wind turbines in Sudan, data from neighbouring countries was used (see Table 2). The analysis was based on the following prices: capital cost = US\$2000/kW, replacement cost = US\$1600/kW, O&M cost = US\$50/kW, as reported by Gebrehiwot et al. [45].

In order to identify the appropriate resources for wind energy (for example, wind speed, ambient temperature, etc.) for the selected RE component, information on the average wind speed and ambient temperature for each month of the year in Sudan was obtained from the database of the National Aeronautics and Space Administration [46]. Fig. 3 illustrates the monthly average wind speed (left axis) and the ambient temperature (right axis). It shows a variable average wind speed across the year. The highest wind speed levels occurred at the beginning and end of the year; in January (7.63 m/s), February (7.84 m/s) and December (7.39 m/s). The lowest values were recorded in September (4.36 m/s) and October (4.49 m/s). Lower wind speeds occur in the warmer months: the average ambient temperature peaked in April (30.62 °C), May (32.05 °C) and June (30.92 °C) and was at a minimum in January (21.91 °C) and December (22.73 °C).

Finally, in setting out the project configuration, the economics were based on a project lifetime of 25 years and defined the nominal discount rate at 6 %, as reported in Elkadeem et al. [15]. In the constraints section, the wind power output was set as 100 % because this study considered only wind energy. The configuration assumed a maximum annual capacity shortage of 100 % because there is no reinforcement of the wind turbine-based system by another energy source. For the optimization configuration, the time step was entered at 60 min in order to simulate hourly generation by the wind energy system for an entire year. Emission penalties were set to zero because Sudan does not levy penalties on emissions. After acquiring the necessary data and then defining the configurations, the HOMER software was run. It simulated the wind turbine-based system's operation, and the results were calculated accordingly.

Concerning the wind turbine performance analysis, HOMER calculates the wind turbine's power output in each time step using a three-step process [47]. First, HOMER calculates the wind speed at the hub height of the wind turbine. Then, in each time step, using Eq. (1), HOMER calculates the wind speed at the hub height of the wind turbine using the inputs that are specified in the Wind Resource page and the Wind Shear entry.

$$U_{hub} = U_{anem} \frac{\ln(Z_{hub}/Z_0)}{\ln(Z_{anem}/Z_0)} \quad (1)$$

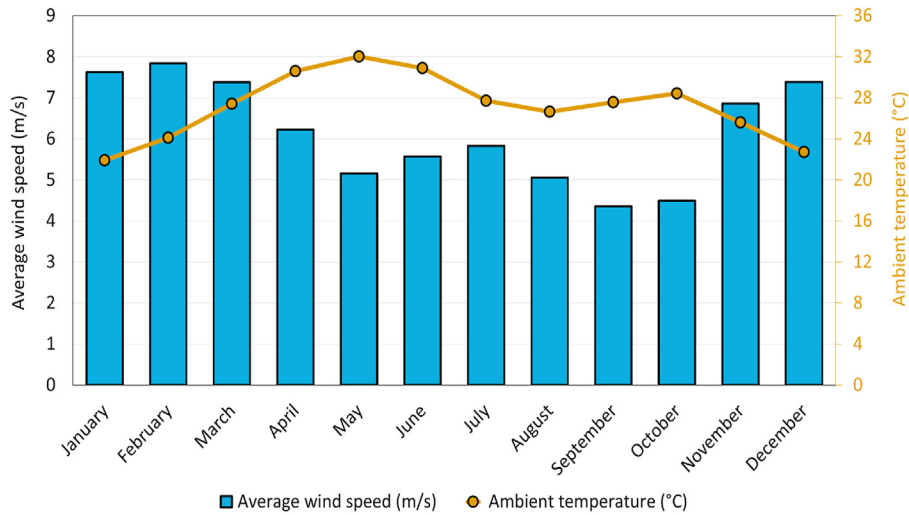
where U_{hub} and U_{anem} are the wind speeds at the hub height (m/s) and anemometer height (m/s), respectively, and Z_{hub} , Z_{anem} and Z_0 are the hub height (m), the anemometer height (m) and the surface roughness length (m), respectively.

Once the hub height wind speed is determined, HOMER then refers to the selected wind turbine's power curve to determine how

Table 2

Capital, replacement, and operational and maintenance (O&M) costs of wind turbine in countries neighbouring Sudan.

Reference	Country	Costs		
		Capital (US\$/kW)	Replacement (US\$/kW)	Operational and maintenance (US\$/kW)
Olatomiwa et al. [41]	Nigeria	6760	4595	25
Maatallah et al. [42]	Tunisia	3977.81	2983.437	77.8125
Diab et al. [43]	Egypt	4500	3000	50
Ramli et al. [44]	Saudi Arabia	3000	1500	40
Gebrehiwot et al. [45]	Ethiopia	2000	1600	50

**Fig. 3.** The average monthly wind speed and ambient temperature profile.

much power the wind turbine produces at that wind speed under standard conditions of temperature and pressure using Eq. (2), as follows:

$$P_{W, stp}(U_{hub}) = \begin{cases} 0 & U_{hub} \leq v_{Wci} \\ \frac{U_{hub} - v_{Wci}}{v_{Wr} - v_{Wci}} \times P_{W, r} & v_{Wci} < U_{hub} \leq v_{Wr} \\ P_{W, r} & v_{Wr} < U_{hub} \leq v_{Wco} \\ 0 & U_{hub} \geq v_{Wco} \end{cases} \quad (2)$$

where $P_{W, stp}$ is the wind turbine power output at the standard values of pressure and temperature, v_{Wci} , v_{Wr} and v_{Wco} are the cut-in, rated and cut-off speed of the chosen wind turbine (m/s), respectively, and $P_{W, r}$ is the wind turbine rated power (kW).

Given that power curves normally specify wind turbine performance under conditions of standard temperature and pressure ($P_{W, stp}$), HOMER multiplies the power value predicted by the power curve by the air density ratio to adjust to actual conditions, according to the following equation (Eq. (3)):

$$P_W = P_{W, stp} \times (\rho / \rho_0) \quad (3)$$

where P_W is the wind turbine output power (kW), ρ_0 is density of the air at the standardised conditions (1.225 kg/m³), and ρ is the actual density of the air (kg/m³).

As for the economic analysis, the cost of energy (COE) per kW and the net present cost (NPC) were the distinguishing features considered among the turbines studied, as suggested by former

techno-economical investigations [15,35,36,47]. The NPC calculates the present cost of installation and operation of the entire system incurred over the project lifetime minus the present revenues, and it can be expressed by Eq. (4), whereas the COE calculates the average cost per kWh of useable electrical energy produced by the system and is calculated utilising Eq. (5), as follows:

$$NPC = \frac{C_{ann, total}}{CRF(i, N)} \quad (4)$$

where $C_{ann, total}$ is the total annualised cost of producing electricity, i is the annual real discount rate, N is the project lifetime, and CRF is the capacity recovery factor, given by $CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1}$

$$COE = \frac{C_{ann, total}}{E_{served}} \quad (5)$$

Further details and sample calculations for the wind turbine performance and economic analyses can be found in Ref. [47]. Fig. 4 depicts a summary of the HOMER operation process flow diagram.

The next phase of the investigation involved simulating the running of a diesel-powered generator rather than a wind turbine system. The intention was to determine the level of emissions that could be avoided by employing wind turbines. Again, the same load profile (20 MW) was assumed. Table 3 summarises the replacement, capital, and O&M costs associated with the diesel generator.

The next step was to simulate the ideal wind turbine (as was identified in the first section of the investigation) operating in the 21 sites. Each site had dissimilar values to the others for wind speed

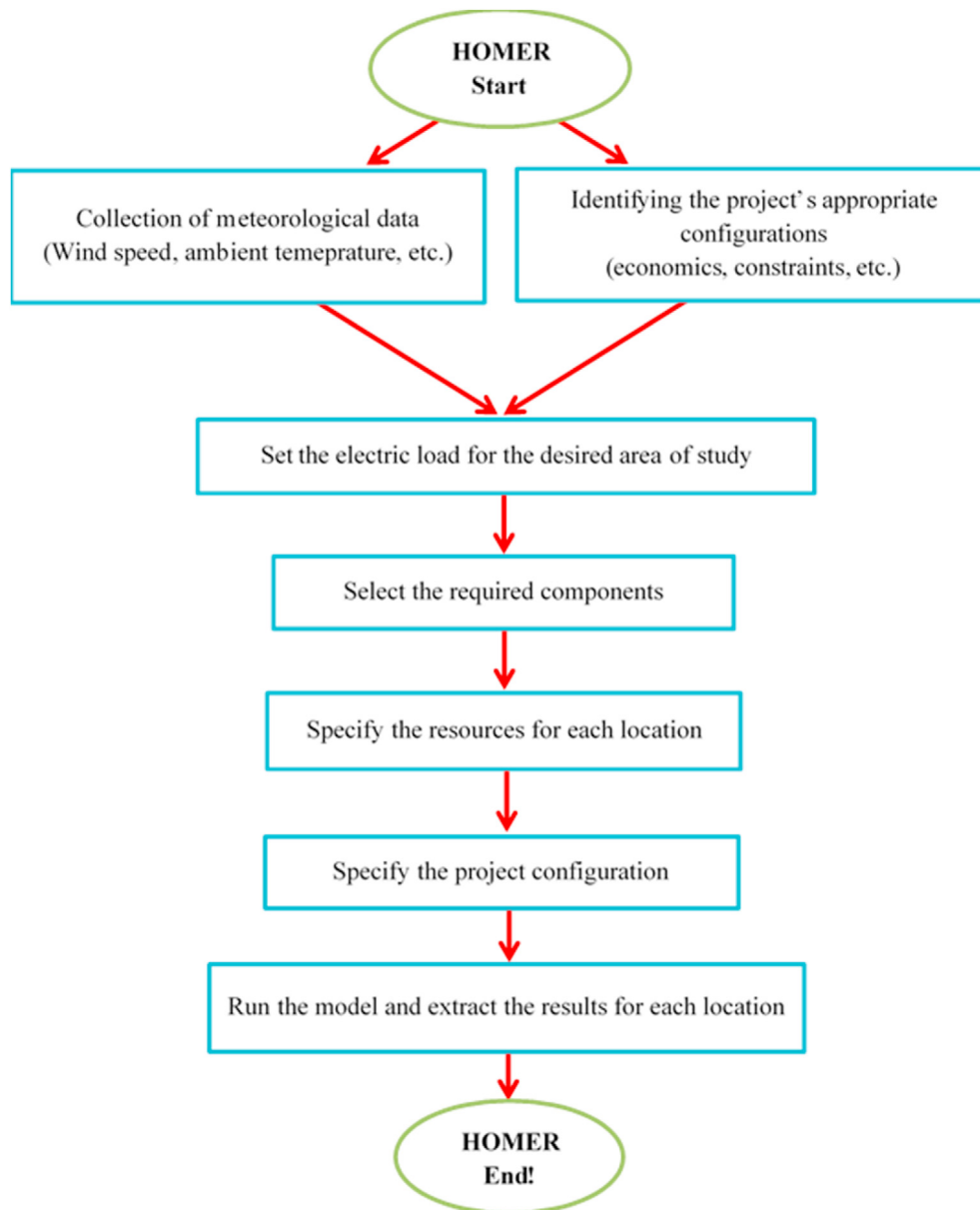


Fig. 4. Process flow diagram of HOMER.

Table 3

Capital, replacement, and operational and maintenance costs for diesel generator [15].

Type	Diesel generator
Capital cost	1000 US\$/kW
Replacement cost	900 US\$/kW
Operation and maintenance cost	0.39 US\$/op. hr
Fuel price [48]	0.159 US\$/L

(Table 4) and ambient temperature (Table 5). However, the electricity load profile and project configurations used were the same as in the study's first phase.

Finally, the impact of wind turbine cost on the economics of the project was explored. This was achieved by altering the cost of wind turbines, taking into account the COE, and using results from the ideal sites that supply the most wind energy.

3. Results and discussion

3.1. Determination of the best wind turbine system

Results for the 25 wind turbine system types with dissimilar capacities are shown in Table 6. It reveals that the net present cost (NPC) for all the turbines was between US\$43.3 M and US\$50.5 M. The HOMER results show that the highest COE was noted for turbine types 9 (Enercon E-82 E4 [3 MW]) and 13 (Enercon E-70 [2.3 MW]), at US\$0.09524/kWh and US\$0.09159/kWh, respectively. The lowest COE was found for types 6 (Enercon E-138 EP3), 25 (Leitwind 80 1000 kW), and 24 (Leitwind 77 1000 kW), at US\$0.05233/kWh, US\$0.0524/kWh, and US\$0.05429/kWh, respectively. The COE varied notably between the studied turbine types, as shown in Fig. 5.

Similarly, the unmet electrical load results varied significantly by wind turbine type (see Fig. 6), ranging between 60.5 % and

Table 4

Monthly average wind speed in m/s for the 21 sites under assessment in Sudan.

Location	January	February	March	April	May	June	July	August	September	October	November	December
Wadi Halfa	6.55	6.70	6.86	6.82	6.99	7.17	6.38	6.76	7.45	7.18	6.63	6.53
Salala	5.34	5.24	5.31	5.25	5.19	5.03	4.28	4.03	4.55	5.13	4.97	5.16
Wawa	7.48	7.52	7.61	7.47	7.71	7.77	6.60	6.94	8.22	8.26	7.67	7.50
Port Sudan	5.95	5.62	5.41	4.80	4.40	4.14	3.89	3.56	3.50	3.96	5.02	5.69
Abu Hamed	7.22	7.33	7.30	7.01	6.75	7.09	6.33	6.26	6.28	6.31	6.77	7.05
Dongola	7.37	7.46	7.54	7.40	7.58	7.60	6.09	6.45	7.78	8.11	7.56	7.37
Karima	7.38	7.54	7.58	7.41	7.22	7.17	5.87	5.86	6.57	7.10	7.20	7.24
Merowe	7.18	7.37	7.42	7.25	6.96	6.78	5.59	5.42	6.04	6.76	6.97	7.00
Al-Goled	7.08	7.25	7.29	7.08	7.08	6.98	5.52	5.61	6.65	7.30	7.13	7.00
Haiya	7.00	6.95	6.84	6.84	6.33	5.13	6.15	5.70	4.87	7.56	7.33	7.24
Ad-Damar	7.32	7.54	7.47	7.03	6.22	5.90	6.01	5.47	4.90	5.60	6.54	7.02
Kassala	6.66	6.90	6.68	5.91	5.04	6.52	6.96	6.06	4.42	3.99	5.62	6.41
Omdurman	7.58	7.83	7.68	7.20	5.96	5.14	6.05	5.66	4.64	5.32	7.24	7.39
Al-Qadarif	7.20	7.47	6.88	5.71	5.00	6.99	6.84	5.84	4.28	3.21	5.06	6.61
Kutum	6.23	6.34	6.57	6.37	5.84	4.78	4.50	4.15	4.78	6.34	6.27	6.14
Al-Fashir	6.36	6.50	6.65	6.27	5.51	4.44	4.31	4.08	4.49	6.01	6.35	6.28
Al-Junaynah	7.02	7.00	6.85	6.04	4.91	4.01	4.26	3.65	3.28	4.81	6.34	6.77
Al-Obeid	7.63	7.83	7.48	6.57	5.34	5.46	5.98	5.25	4.49	4.79	7.11	7.46
Kosti	7.47	7.78	7.30	6.10	5.37	6.62	6.33	5.47	4.69	4.21	6.05	7.11
Al-Damazin	6.20	6.41	5.88	5.41	5.58	6.33	5.72	4.98	3.99	3.26	3.96	5.51
Nyala	7.99	8.04	7.77	6.82	5.26	4.67	4.84	4.21	4.10	5.38	7.38	7.80

Table 5

Monthly average ambient temperature in °C for the 21 sites under assessment in Sudan.

Location	January	February	March	April	May	June	July	August	September	October	November	December
Wadi Halfa	15.14	17.09	21.41	26.77	30.83	32.31	33.09	33.25	31.33	27.56	21.21	16.51
Salala	18.50	19.69	22.61	26.85	30.76	33.22	34.29	34.21	32.35	28.42	23.93	20.26
Wawa	15.98	18.00	22.38	27.64	31.64	33.06	33.89	34.20	32.32	28.30	21.85	17.25
Port Sudan	22.55	22.72	24.21	26.79	29.96	32.34	33.48	33.48	31.94	29.17	26.66	24.16
Abu Hamed	18.51	20.21	24.06	28.93	32.90	34.48	34.77	35.01	34.20	30.76	24.88	20.25
Dongola	17.17	19.23	23.56	28.69	32.64	34.02	34.77	35.13	33.47	29.50	22.98	18.42
Karima	18.31	20.20	24.18	29.07	32.92	34.36	34.66	34.90	34.04	30.60	24.35	19.82
Merowe	18.78	20.70	24.64	29.46	33.22	34.57	34.71	34.86	34.30	31.07	24.77	20.28
Al-Goled	18.11	20.17	24.31	29.22	33.02	34.38	34.82	35.04	34.04	30.42	23.90	19.41
Haiya	21.57	22.70	25.32	28.89	32.13	33.88	32.17	31.03	31.54	28.90	25.60	22.86
Ad-Damar	20.87	22.52	25.98	30.40	33.88	35.28	34.39	33.96	34.27	31.76	26.78	22.58
Kassala	25.45	26.89	29.40	32.50	33.25	31.75	27.78	26.30	27.27	28.89	27.95	26.22
Omdurman	21.21	23.15	26.65	30.70	33.75	34.36	32.42	31.39	32.42	31.75	26.54	22.53
Al-Qadarif	25.45	26.99	29.56	32.28	32.40	30.70	27.31	26.17	26.93	28.83	28.05	26.19
Kutum	17.87	20.46	23.67	26.90	28.86	29.36	27.49	26.27	26.64	25.22	21.22	18.27
Al-Fashir	18.91	21.51	24.82	28.02	29.99	30.47	28.57	27.31	27.75	26.55	22.50	19.39
Al-Junaynah	22.11	24.59	27.55	30.08	30.83	29.56	26.26	24.56	25.27	25.58	24.03	22.20
Al-Obeid	21.30	23.55	26.98	30.39	32.33	31.82	28.77	27.62	28.60	29.17	25.47	22.20
Kosti	24.97	26.86	29.82	32.72	33.34	31.98	28.76	27.60	28.45	30.38	28.77	25.97
Al-Damazin	26.62	28.30	30.64	32.18	30.87	28.84	26.45	25.85	26.77	28.40	28.73	27.09
Nyala	21.83	24.27	27.28	29.85	30.75	29.80	26.77	25.73	26.42	26.80	24.79	22.31

78.1 %.

O&M expenses depend strongly on how many wind turbine units are required to generate enough electricity to source the targeted demand of 20 MW. The least number of units needed was with type 1 (Enercon E–126 [7.5 MW]), whereas the highest was with type 25 (Leitwind 80 1000 kW).

Based on these results, the ideal wind turbine for a town in Sudan could be identified. It had the lowest COE (about US\$0.05233/kWh), and only six units were required to meet a 20 MW demand. It was identified as type 6 (Enercon E–138 EP3); its properties are summarised in Table 7. This type of wind turbine was thus chosen for the subsequent analyses.

3.2. Wind turbines vs diesel-powered generator systems

3.2.1. Economic assessment

The assessment showed that the COE for the diesel-powered generator system was considerably more expensive at US\$0.9655/kWh than the US\$0.05233/kWh for energy from wind turbines.

Thus, implementing a wind turbine-based system will save around US\$0.91317/kWh.

However, the analysis assumed a fixed price value for the diesel generator (capital cost = US\$1000/kW, replacement cost = US\$900/kW) and its diesel fuel (US\$0.159/L). Still, in reality, these prices will vary throughout the project lifetime. For example, the emergence of the Coronavirus epidemic significantly lowered global energy demand and consequently led to a reduction in fossil fuel prices [49]. The effect is a forecast decline in diesel generator prices. Such a decrease might negatively impact the competitiveness of RE projects, particularly wind energy projects, endangering the achievement of several Sustainable Development Goals and the COP25 Paris Agreement.

In recognition of likely future instabilities of diesel generator and diesel fuel prices, a sensitivity analysis was conducted, and the prices of diesel generator and diesel fuel were altered to investigate their influence on the COE expense difference. Fig. 7 shows how price reductions for diesel generators and their fuel impact the diesel-powered generator system COE. The results reveal that even

Table 6
HOMER simulation results for various types of wind turbine systems.

No.	Wind turbine type	Capacity (kW)	Hub height (m)	COE (US\$/kWh)	NPC (US\$)	Unmet electric load (%)	No. of wind turbines essential to attain the desired demand of 20 MW ($\frac{20 \text{ MW}}{\text{Capacity of the selected wind turbine type}}$)
1	Enercon E-126 [7.5 MW]	7580	135	0.08851	50.5 M	74.5	3
2	Enercon E-115 EP3 E3	4200	122	0.07156	46.6 M	70.9	5
3	Enercon E-126 EP4 [4.2 MW]	4200	135	0.06304	46.6 M	67	5
4	Enercon E-126 EP3	4000	135	0.06195	44.4 M	68	5
5	Enercon E-101 E2 [3.5 MW]	3500	74	0.08145	46.6 M	74.5	6
6	Enercon E-138 EP3	3500	135	0.05233	46.6 M	60.5	6
7	Enercon E-115 E2 3.2 MW	3200	122	0.05885	42.6 M	67.7	6
8	Enercon E-115 [3 MW]	3000	122	0.05744	46.6 M	63.8	7
9	Enercon E-82 E4 [3 MW]	3000	84	0.09524	46.6 M	78.1	7
10	Leitwind 101 3000 kW	3000	143	0.06506	46.6 M	68	7
11	Leitwind 101 2500 kW	2500	94	0.06273	44.4 M	68.4	8
12	Enercon E-103 EP2	2350	74	0.0635	46.9 M	67	9
13	Enercon E-70 [2.3 MW]	2300	85	0.09159	45.9 M	77.6	9
14	Enercon E-82 E2 [2 MW]	2000	85	0.06869	44.4 M	71.1	10
15	Leitwind 101 2000 kW	2000	80	0.05642	44.4 M	64.9	10
16	Leitwind 90 2000 kW	2000	98	0.06295	44.4 M	68.5	10
17	Leitwind 80 1800 kW	1800	80	0.07463	43.9 M	73.7	11
18	Leitwind 80 1650 kW	1650	65	0.07267	43.9 M	73	12
19	Vestas V82 [1.65 MW]	1650	70	0.07291	43.9 M	73.1	12
20	Leitwind 77 1500 kW	1500	65	0.07376	43.3 M	73.8	13
21	Leitwind 80 1500 kW	1500	60	0.06923	43.3 M	72.1	13
22	Leitwind 86 1500 kW	1500	100	0.05934	43.3 M	67.4	13
23	Leitwind 90 1500 kW	1500	80	0.05536	43.3 M	65.1	13
24	Leitwind 77 1000 kW	1000	80	0.05429	44.4 M	63.5	20
25	Leitwind 80 1000 kW	1000	80	0.0524	44.4 M	62.2	20

COE = cost of energy; NPC = net present cost.

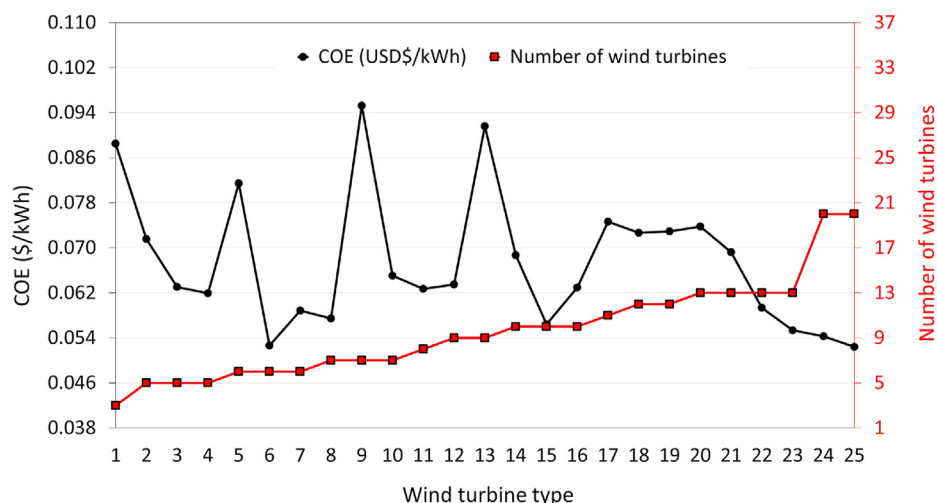


Fig. 5. Cost of energy (COE) of the examined wind turbines. 1 Enercon E-126 [7.5 MW]. 2 Enercon E-115 EP3 E3. 3 Enercon E-126 EP4 [4.2 MW]. 4 Enercon E-126 EP3. 5 Enercon E-101 E2 [3.5 MW]. 6 Enercon E-138 EP3. 7 Enercon E-115 E2 3.2 MW 8 Enercon E-115 [3 MW]. 9 Enercon E-82 E4 [3 MW]. 10 Leitwind 101 3000 kW 11 Leitwind 101 2500 kW 12 Enercon E-103 EP2. 13 Enercon E-70 [2.3 MW]. 14 Enercon E-82 E2 [2 MW]. 15 Leitwind 101 2000 kW 16 Leitwind 90 2000 kW 17 Leitwind 80 1800 kW 18 Leitwind 80 1650 kW 19 Vestas V82 [1.65 MW]. 20 Leitwind 77 1500 kW 21 Leitwind 80 1500 kW 22 Leitwind 86 1500 kW 23 Leitwind 90 1500 kW 24 Leitwind 77 1000 kW 25 Leitwind 80 1000 kW.

in the face of such price reduction, the economic performance of the ideal wind turbine system remains superior to the typical diesel-powered generator system. This demonstrates that the endorsed system is economically viable and sustainable even in the face of unstable prices for diesel generators and their fuel.

3.2.2. Emissions of greenhouse gases (GHGs)

Like all other forms of RE, wind turbines are considered clean and environmentally friendly because they generate zero emissions, unlike fossil fuels, which produce substantial quantities of pollutants. Accordingly, the operation of wind turbines averts the

emissions that result from the combustion of fossil fuel.

Diesel is the fossil fuel that is primarily used in Sudan's in off-grid electricity plants. The combustion products from diesel have numerous adverse impacts on both human health and the environment. The impacts of diesel particulates on humans include respiratory complications and the hindering of oxygen delivery to body tissues. Environmentally, diesel emissions result in atmospheric reactions that lead to photochemical smog and acid rain [43]. Table 8 lists the pollutants (carbon dioxide, carbon monoxide, nitrogen oxide, sulfur dioxide, unburned hydrocarbons and particulate matter) that a diesel generator expels when it operates

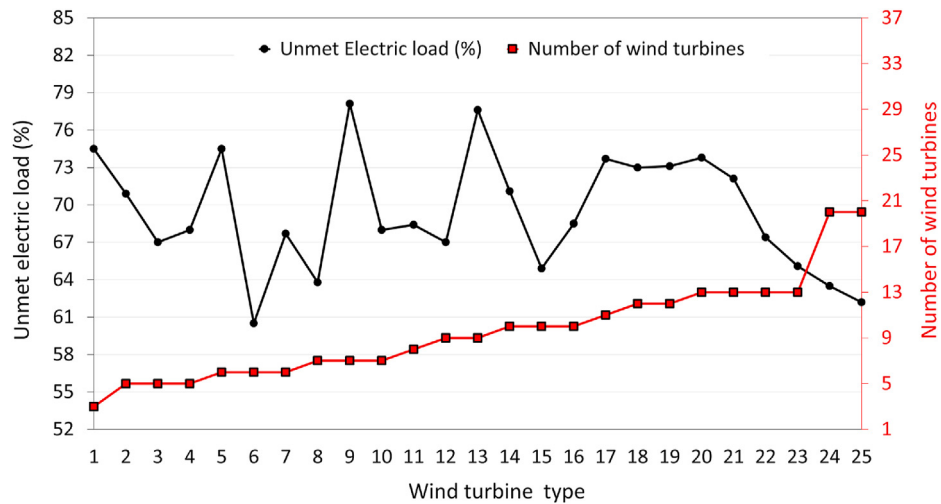


Fig. 6. The unmet electrical load of the examined wind turbines. 1 Enercon E–126 [7.5 MW], 2 Enercon E–115 EP3 E3, 3 Enercon E–126 EP4 [4.2 MW], 4 Enercon E–126 EP3, 5 Enercon E–101 E2 [3.5 MW], 6 Enercon E–138 EP3, 7 Enercon E–115 E2 3.2 MW 8 Enercon E–115 [3 MW], 9 Enercon E–82 E4 [3 MW], 10 Leitwind 101 3000 kW 11 Leitwind 101 2500 kW 12 Enercon E–103 EP2, 13 Enercon E–70 [2.3 MW], 14 Enercon E–82 E2 [2 MW], 15 Leitwind 101 2000 kW 16 Leitwind 90 2000 kW 17 Leitwind 80 1800 kW 18 Leitwind 80 1650 kW 19 Vestas V82 [1.65 MW], 20 Leitwind 77 1500 kW 21 Leitwind 80 1500 kW 22 Leitwind 86 1500 kW 23 Leitwind 90 1500 kW 24 Leitwind 77 1000 kW 25 Leitwind 80 1000 kW.

Table 7

Properties of the optimal wind turbine system.

Name	Enercon E–138 EP3
Rated capacity	3500 kW
Number of blades	3
Hub height	135 m
Rotor type	Upwind rotor with active pitch control
Rotor diameter	127 m
Blade material	Glass-reinforced plastic (GRP) (epoxy resin); Built-in lightning protection
Cut-out wind speed	28–34 m/s with ENERCON storm control

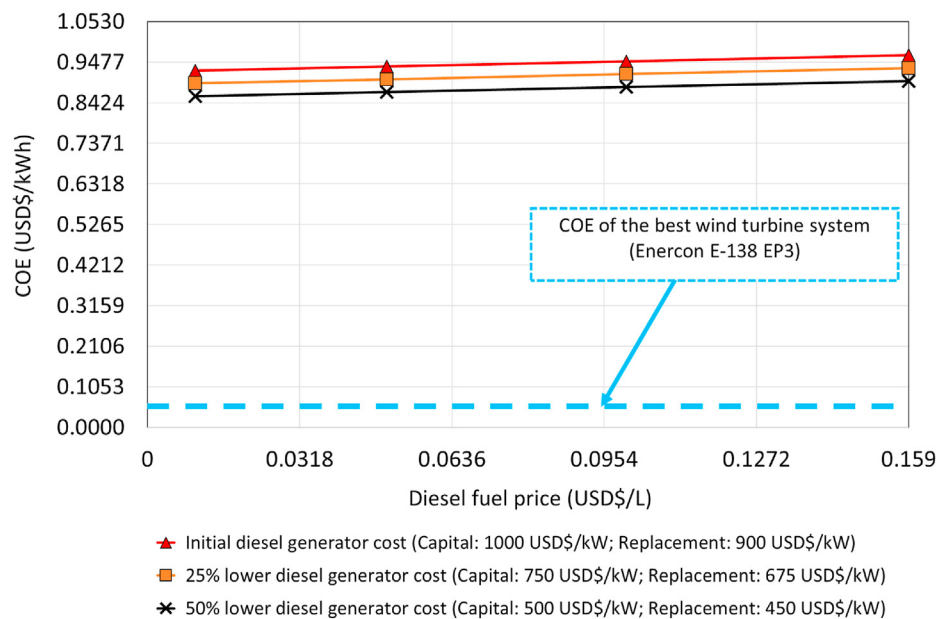


Fig. 7. Diesel generator and fuel price reduction impact on the COE of the diesel-powered generator system.

with a met electric load percentage of 39.5 % (which is close to the met electrical load of an optimal wind turbine). The HOMER results

demonstrate that the employment of wind turbines in Sudan would avoid 48,891,635 kg of pollutants being generated annually.

Table 8

Greenhouse gas emissions by a diesel generator with a met electric load percentage of 39.5 %.

Quantity	Emissions (kg/year)
Carbon dioxide (CO ₂)	48,169,721
Carbon monoxide (CO)	303,635
Nitrogen oxides (NO _x)	285,233
Sulfur dioxide (SO ₂)	117,956
Unburned hydrocarbons (UHC)	13,250
Particulate matter (PM)	1840
Total emissions	48,891,635

3.3. Evaluation of the performance of the best wind turbine system

Table 9 depicts HOMER simulation results employing the optimal wind turbine system, the Enercon E–138 EP3, for the 21 assessed sites assessed. The lowest COE was obtained in Wawa (US\$0.0409/kWh) and the highest in Port Sudan at US\$0.08438/kWh. Fig. 8 lists the COE for each site, and the four lowest-COE sites are emphasised in red. The unmet electric load ranged from 49.1 to 75.3 %; Wawa had the lowest percentage, and Port Sudan had the highest (see Fig. 9; the four lowest unmet load sites are highlighted in red).

High average wind speeds characterise these regions, and wind turbine-based systems are ideal for generating electricity in remote areas. The fuel for a wind turbine-based power station is the local

Table 9

HOMER simulation results utilising the optimal wind turbine (Enercon E–138 EP3) for the 21 sites under assessment in Sudan.

Location	COE (US\$/kWh)	NPC (US\$)	OC (US\$)	Unmet electric load (%)
Wadi Halfa	0.04502	46.6 M	0.36 M	53.8
Salala	0.07330	46.6 M	0.36 M	71.6
Wawa	0.04090	46.6 M	0.36 M	49.1
Port Sudan	0.08438	46.6 M	0.36 M	75.3
Abu Hamed	0.04557	46.6 M	0.36 M	54.3
Dongola	0.04211	46.6 M	0.36 M	50.6
Karima	0.04429	46.6 M	0.36 M	53
Merowe	0.04649	46.6 M	0.36 M	55.2
Al-Goled	0.04556	46.6 M	0.36 M	54.3
Haiya	0.04811	46.6 M	0.36 M	56.7
Ad-Damar	0.0491	46.6 M	0.36 M	57.6
Kassala	0.05496	46.6 M	0.36 M	62.1
Omdurman	0.04899	46.6 M	0.36 M	57.5
Al-Qadarif	0.05537	46.6 M	0.36 M	62.4
Kutum	0.05767	46.6 M	0.36 M	63.9
Al-Fashir	0.05955	46.6 M	0.36 M	65.1
Al-Junaynah	0.06421	46.6 M	0.36 M	67.6
Al-Obeid	0.05104	46.6 M	0.36 M	59.2
Kosti	0.05193	46.6 M	0.36 M	59.9
Al-Damazin	0.06674	46.6 M	0.36 M	68.8
Nyala	0.05258	46.6 M	0.36 M	60.4

COE = cost of energy; NPC = net present cost; OC = operating cost.

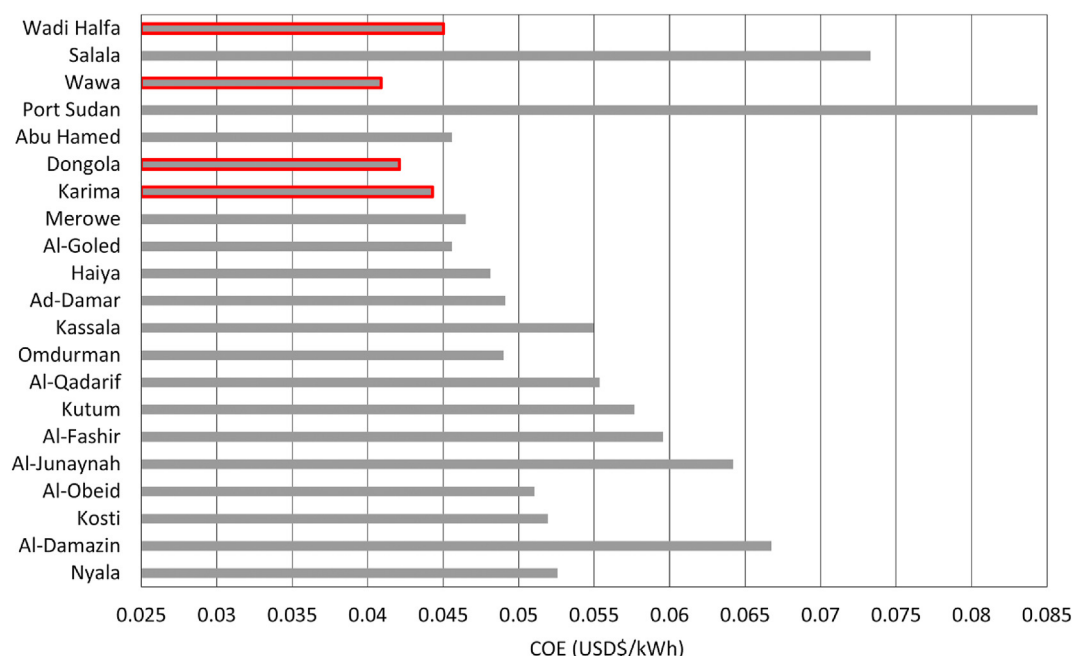


Fig. 8. The cost of energy (COE) for the 21 sites under assessment in Sudan.

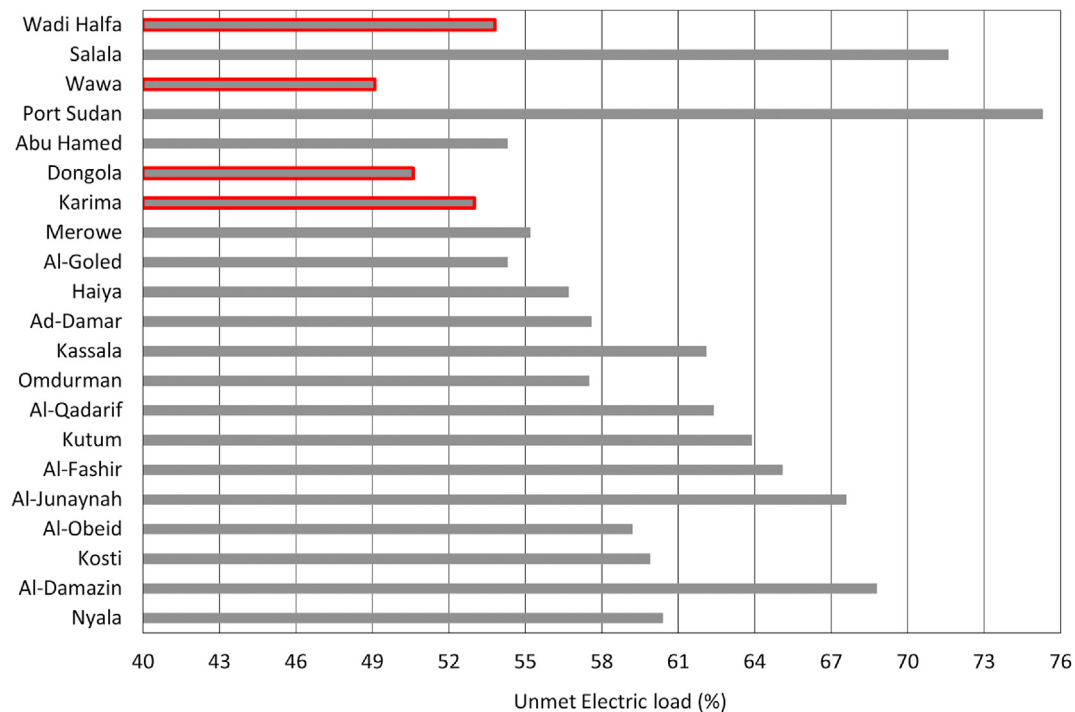


Fig. 9. The unmet electrical load for the 21 locations under assessment in Sudan.

wind resource, and even minor variations significantly impact a wind farm's commercial value. Slight changes in average speed can substantially alter the turbine's performance [50]. For instance, a doubling of the average wind speed increases the wind's power by a factor of eight. Similarly, an increase in a site's average wind speed from 6 m/s to 10 m/s will escalate the energy produced by a wind farm by over 130 % [50].

Another parameter that strongly influences energy production from wind turbines is air density. The power available from the wind (i.e. the pressure exerted on wind turbine blades) correlates directly with air density: as air density increases, the available power also increases [51,52]. Furthermore, air density depends strongly on the ambient temperature [53]: as ambient temperature increases, the air density decreases, which profoundly reduces the power available in the wind and therefore adversely affects the amount of energy generated.

One other crucial factor in wind turbine system performance is terrain-induced turbulence. When winds flow over regions with irregular topographies (i.e., mountains, hills, creeks, etc.), they often change direction and speed suddenly. These sudden changes (turbulence), depending on their intensity, may undesirably affect wind turbines' mechanical components and wind farms' economic performance [42]. Thus turbulence intensity has an inverse relationship with a wind turbine's power output, according to Pagnini et al. [54]. Therefore, wind turbines should be deployed in regions with less complex terrain that are endowed with strong winds and high air density for the best wind energy production.

Out of the 21 sites analysed, the results clearly show that the best sites for wind energy production are Wawa, Dongola, Karima, Wadi Halfa, and Al-Goled. These locations comprise an arid desert with a less complex topography and a relatively low turbulence intensity, and they have the highest wind levels throughout the year (Fig. 10). Wawa and Dongola are the sites of Sudan's maximum wind speed levels, followed by Karima, Wadi Halfa and Al-Goled (Fig. 10). Accordingly, the results show these locations to have the lowest COE compared to the other sites. Cost-wise, Wawa's COE

was US\$0.0409/kWh, followed by Dongola at US\$0.04211/kWh, Karima at US\$0.04429/kWh, Wadi Halfa at US\$0.04502/kWh, and Al-Goled at US\$0.04556/kWh. Accordingly, there is significant potential for thorough exploitation of wind energy at these sites, particularly at locations such as Wawa, Dongola and Karima.

The average monthly electricity production for these three sites is shown in Fig. 11, which shows that the highest generation levels occur at the beginning and end of the year, with extra peaks during March and October. In these locations, wind speed is much higher during these months compared to the remaining months, and generation levels are elevated accordingly at this time. Generation tends to decline from May until July, reflecting the effects of warmer temperatures. As previously mentioned, temperature affects the power output of wind turbine-based systems; high temperatures reduce the air density and, consequently, the wind turbine's power output. As Fig. 11 shows, the air temperature rises sharply as summer approaches (from late April to late July). This causes decreased air density around the turbines, causing their efficiency to decline and lower electricity generation levels.

In contrast, many factors make Port Sudan, Salala, Al-Damazin, Al-Junaynah and Al-Fashir less appealing for wind energy generation. As coastal towns, Port Sudan and Salalah have exceptionally high levels of humidity. Exposure to high humidity during their 25-year lifetime makes wind turbines susceptible to severe tribological conditions such as corrosion, erosion, delamination and other hazardous degradation processes. This can increase damage to components and cause the malfunction of machines [55–59]. These factors increase the maintenance cost. Furthermore, the mountainous topography that defines these locations increases the intensity of terrain-induced turbulence, causing undesirable consequences on turbines' mechanical parts and their power output. Concerning wind availability (Fig. 10), Port Sudan has Sudan's lowest wind speeds, followed by Salala, Al-Damazin, Al-Junaynah and Al-Fashir. Accordingly, these sites have the most expensive COE values of all the sites. The respective COEs were: Port Sudan US\$0.08438/kWh; Salala US\$0.07330/kWh; Al-Damazin

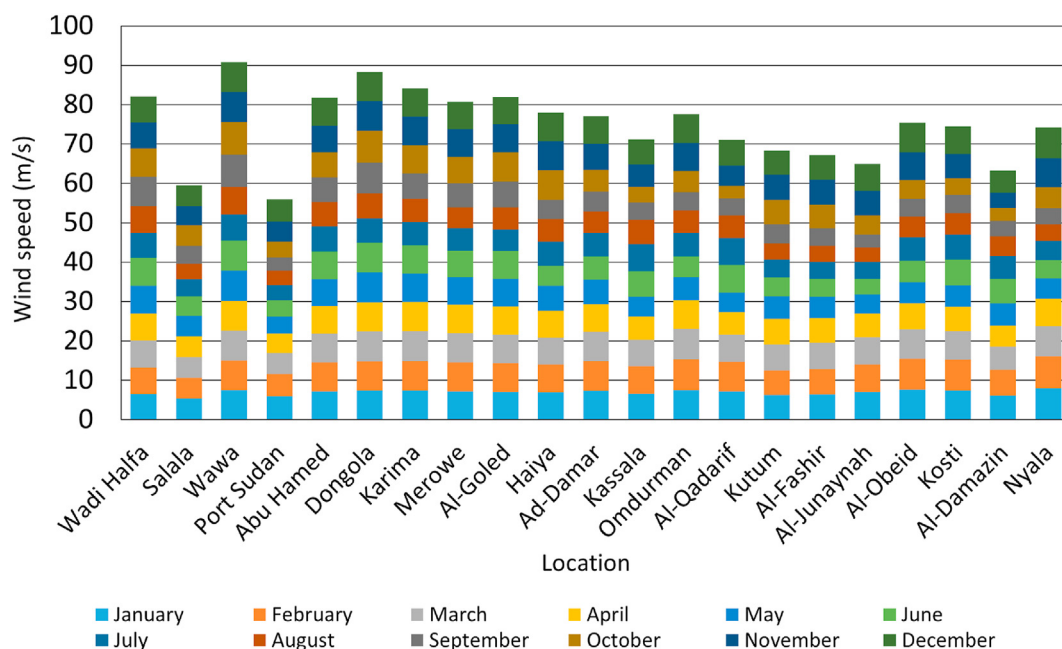


Fig. 10. Wind speed values for the 21 sites under assessment in Sudan.

US\$0.06674/kWh, Al-Junaynah US\$0.06421/kWh and Al-Fashir US\$0.05955/kWh.

3.4. Influence of wind turbine cost on the economics of the renewable energy project

Because of the predicted ongoing global decline in wind turbine pricing, the turbine cost was altered in order to understand its impact on the project economics. Fig. 12 shows how wind turbine cost alterations influence COE; data are shown for the four sites with the lowest COE. A decrease in the cost of wind turbines by approximately 10 % (capital cost = US\$1800/kW, replacement cost = US\$1440/kW) has a substantial impact, reducing the COE generated from these wind turbine plants to US\$0.03681/kWh, US\$0.0379/kWh, US\$0.03986/kWh, and US\$0.04052/kWh for Wawa, Dongola, Karima and Wadi Halfa, respectively. A further reduction in the cost of turbines of up to 25 % radically diminishes the COE to between US\$0.03067/kWh and US\$0.03376/kWh at the top four sites. Therefore, the predicted ongoing reduction in turbine costs will make the opportunity for wind turbines to contribute to Sudan's energy mix even more attractive fundamentally.

4. Conclusion

This investigation used the HOMER modelling framework to identify the ideal type of wind turbine to employ for a town in Sudan to meet an electric load of 20 MW. The Enercon E-138 EP3 turbine was identified as the optimum type because it had the lowest COE of the turbines examined. A relatively low number of wind turbine units were required during the simulation. The results of the simulation demonstrated that establishing a wind turbine-based system will save almost US\$0.91317/kWh on COE compared to a diesel generator system. In addition, a reduction in pollutants of nearly 49 million kg/year can also be achieved.

To investigate which locations in Sudan are most suitable for wind energy generation, the ideal wind turbine system was simulated to run in 21 dissimilar sites. Four sites stood out for their high wind speed records and low COE: Wawa, Dongola, Karima, Wadi

Halfa, and Al-Goled, in that order. For these sites, the economic effect of the predicted ongoing rapid decrease in wind turbine price was then investigated by altering the simulation's price variable. This showed that an approximate 10 % decrease in wind turbine costs lowered the COE to between US\$0.03681/kWh and US\$0.04052/kWh. A further turbine price decrease of up to 25 % markedly diminished the COE to between US\$0.03067/kWh and US\$0.03376/kWh.

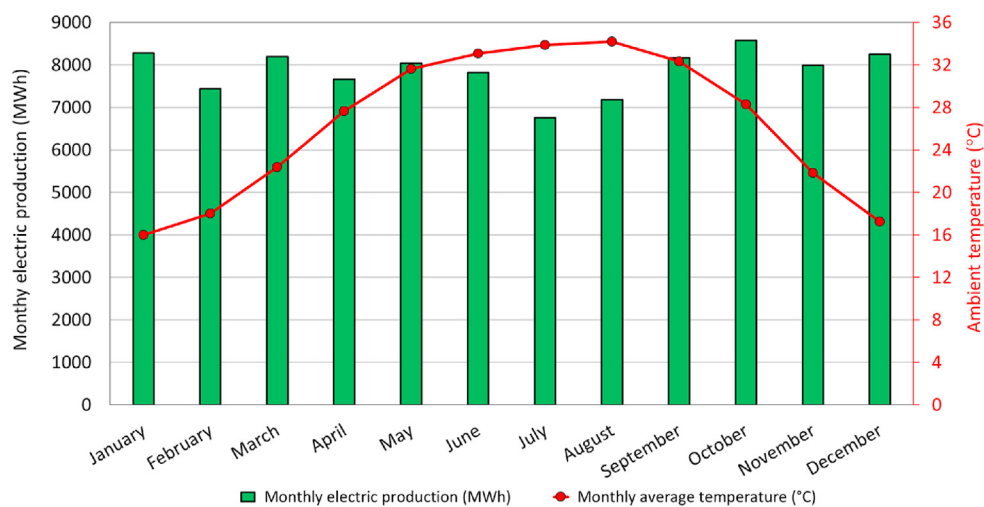
In view of the above, the establishment of wind farms is feasible and will constitute enormous progress in Sudan's RE sector, primarily by adding energy generation capacity while increasing energy security and addressing both national and global environmental concerns. The findings of this investigation provide a strong impetus for investors and policymakers to invest more in wind energy in Sudan. This will provide employment opportunities that will mitigate poverty and offer practical steps towards achieving an energy future for Sudanese people that is cleaner, more sustainable and more secure than the current inadequate system. Finally, Sudan can become a vital exporter in the subregion, providing significant economic gain and strengthening regional integration.

CRediT authorship contribution statement

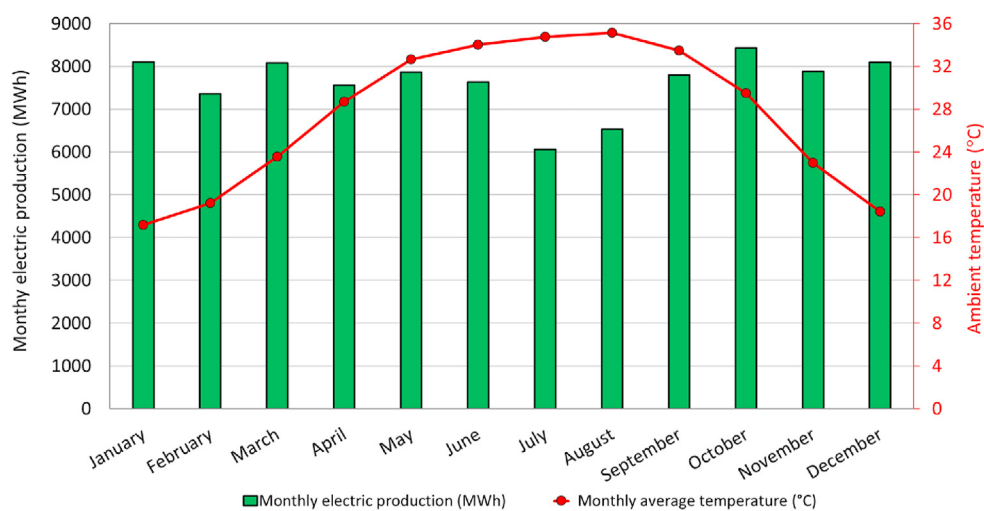
Sulaiman O. Fadlallah: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Supervision, Project administration, Funding acquisition. **Djamal Eddine Benhadji Serradj:** Methodology, Software, Formal analysis, Investigation, Resources, Visualisation, Writing – original draft. **Delight M. Sedzro:** Methodology, Formal analysis, Investigation, Data curation, Writing – review & editing, Funding acquisition.

Declaration of competing interest

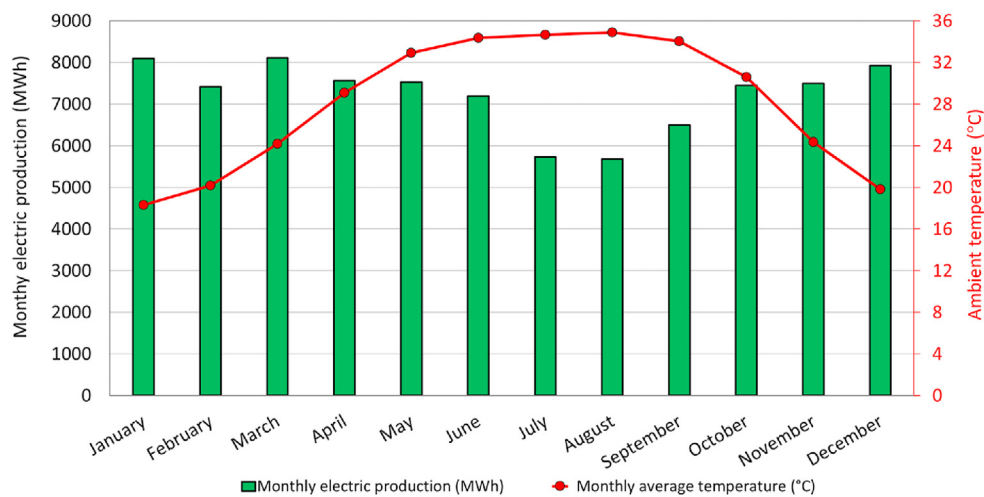
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



(a) Wawa



(b) Dongola



(c) Karima

Fig. 11. Monthly average electric production and ambient temperature profile for Wawa, Dongola and Karima.

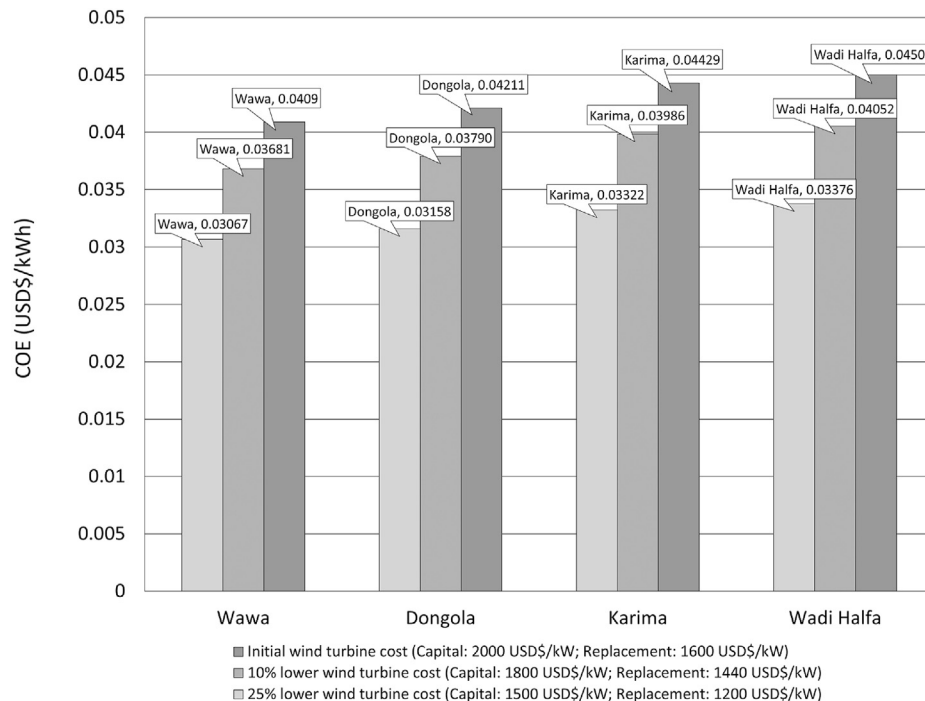


Fig. 12. Impact of wind turbine cost on the COE for the top four locations in Sudan.

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